10. 试证:

(1) 与所有 n 阶对角阵乘法可交换的矩阵也必是 n 阶对角阵;

(2) 与所有 n 阶矩阵乘法可交换的矩阵是纯量阵.

11. 证明: 两个对角元为 1 的上三角阵乘积仍是对角元为 1 的上三角阵.

证: (1) 设 $A_n = \mathrm{diag}(a_1 \cdots a_n)$

$B_{n\times n} = [b_{ij}]_{n\times n}$

$AB = BA$

$\mathrm{entry}(AB, i, j) = \mathrm{entry}(BA, i, j)$

$a_j b_{ij} = b_{ij} a_i$

$i \neq j$ 时 $a_i \neq a_j$ 则 $b_{ij} = 0$

$\therefore B$ 非主对角线元素均为 0

(2) $A_{n\times n} = [a_{ij}]_{n\times n}$ $B_{n\times n} = [b_{ij}]_{n\times n}$

$\mathrm{row}(A, i)\,\mathrm{col}(B, j) = \mathrm{row}(B, i)\,\mathrm{col}(A, j)$

遍取只有 1 个元素为 1 其他均为 0 的矩阵

$AE_{ij} = E_{ij}A$

$c_{ii} = \sum_{k=1}^{n} a_{ik} b_{ki} = \sum_{k=1}^{n} b_{ik} a_{ki}$

$a_{i1} b_{1i} \neq b_{i1} a_{1i}$

$a_{i2} b_{2i} \neq b_{i2} a_{2i}$

$b_{1i} = b_{i1} = \cdots = 0$

$a_{ii} b_{ii} = a_{ii} b_{ii}$

15. 设 n 阶方阵 $\boldsymbol{A},\boldsymbol{B}$ 满足 $\boldsymbol{A}^2=\boldsymbol{A},\boldsymbol{B}^2=\boldsymbol{B}$, 且 $(\boldsymbol{A}+\boldsymbol{B})^2=\boldsymbol{A}+\boldsymbol{B}$, 证明: $\boldsymbol{AB}=\boldsymbol{O}$.

16. 设 $\boldsymbol{A}=\begin{bmatrix}1&0&1\\0&2&0\\1&0&1\end{bmatrix}$, 求 $\boldsymbol{A}^n-2\boldsymbol{A}^{n-1}(n\geqslant 2)$.

17. 设 n 阶方阵 $\boldsymbol{A},\boldsymbol{B}$ 满足 $\boldsymbol{A}^2=-\boldsymbol{A},\boldsymbol{B}^2=-\boldsymbol{B}$, 且 $(\boldsymbol{A}+\boldsymbol{B})^2=-\boldsymbol{A}-\boldsymbol{B}$, 证明: $\boldsymbol{AB}=\boldsymbol{O}$

15.17. $AB+BA=0$

$AB+BAB=0$

$BAB+BA=0$

$\therefore AB=BA$

20. 设 n 维向量 $\boldsymbol{\alpha}=[a,0,\cdots,0,a]^{\mathrm{T}}$, $a<0$, $\boldsymbol{I}$ 为 n 阶单位矩阵, 矩阵

$$\boldsymbol{A}=\boldsymbol{I}-\boldsymbol{\alpha}\boldsymbol{\alpha}^{\mathrm{T}},\quad \boldsymbol{B}=\boldsymbol{I}+\frac{1}{a}\boldsymbol{\alpha}\boldsymbol{\alpha}^{\mathrm{T}}$$

其中 $\boldsymbol{AB}=\boldsymbol{I}$, $\boldsymbol{I}$ 为 n 阶单位矩阵, 求 $\boldsymbol{\alpha}$.

敏感结合!

$AB=I+\frac{1}{a}\alpha\alpha^T-\alpha\alpha^T-\frac{1}{a}\alpha\alpha^T\alpha\alpha^T=I$

$(\frac{1}{a}-1)\alpha\alpha^T-2a\,\alpha\alpha^T=0$

$(\frac{1}{a}-1-2a)\alpha\alpha^T=0$

$\frac{1}{a}-1-2a=0$

$\therefore a=-1$

30. 设 $\boldsymbol{A}$ 为 $m \times n$ 矩阵, 且 $\operatorname{rank}(\boldsymbol{A}) = r$, 从 $\boldsymbol{A}$ 中任取 s 行构成一个 $s \times n$ 矩阵 $\boldsymbol{B}$, 证明: $\operatorname{rank}(\boldsymbol{B}) \geqslant r + s - m$.

30. 设对A作初等行变换，使之变成 $\begin{bmatrix} B \\ C \end{bmatrix}$

设 ~~Rank(B)~~ 则 $r(B) \leq s$ $\quad r(C) \leq m-s$

$r(B)+r(C)=r$

$\therefore r(B) \geq r+s-m$

例1.2.5 设 n 阶矩阵 $\boldsymbol{A}$ 和 $\boldsymbol{B}$ 的每列元素之和均为1,

(1) 计算 $\mathbf{1}^{\mathrm{T}}\boldsymbol{A}$ 和 $\mathbf{1}^{\mathrm{T}}\boldsymbol{B}$, 其中 $\mathbf{1}$ 为 n 元列向量 $\mathbf{1} = [1, 1, \cdots, 1]^{\mathrm{T}}$;

(2) 证明矩阵 $\boldsymbol{AB}$ 的每列元素之和均为 1.

(1) 1^TA 为 $1\times n$ 矩阵

$1^T col(A,j) = \sum_{i=1}^{n} a_{ij} = 1$

$\therefore 1^TA = 1^T$

(2) AB第j列元素之和为

牛! $1^T col(AB,j) = 1^TA\,col(B,j) = (1^TA)\,col(B,j) = 1^T col(B,j) = 1$

4. 讨论下列线性方程组，当 λ 取什么值时方程组有唯一解？取什么值时有无穷多解？取什么值时无解？

(1)

$$\begin{cases}(\lambda+3)x_1+x_2+2x_3=\lambda\\ \lambda x_1+(\lambda-1)x_2+x_3=\lambda\\ 3(\lambda+1)x_1+\lambda x_2+(\lambda+3)x_3=3\end{cases}$$

此类题形式简单者初变复杂者行列式

$$|A|=\begin{vmatrix}\lambda+3 & 1 & 2\\ \lambda & \lambda-1 & 1\\ 3(\lambda+1) & \lambda & \lambda+3\end{vmatrix}=(\lambda+3)^2(\lambda-1)+3(\lambda+1)+2\lambda^2-6(\lambda^2-1)-\lambda(\lambda+3)-\lambda(\lambda+3)$$

$$=(\lambda^2+6\lambda+9)(\lambda-1)+3\lambda+3+2\lambda^2-6\lambda^2+6-\lambda^2-3\lambda-\lambda^2-3\lambda$$

$$=\lambda^3-\lambda^2+6\lambda^2-6\lambda+9\lambda-9+3\lambda+3+2\lambda^2-6\lambda^2+6-\lambda^2-3\lambda-\lambda^2-3\lambda$$

$$=\lambda^2(\lambda-1)$$

① $\lambda=0$

增广阵 $\overline{A}=\begin{pmatrix}3 & 1 & 2 & 0\\ 0 & -1 & 1 & 0\\ 3 & 0 & 3 & 3\end{pmatrix}\to\begin{pmatrix}1 & 0 & 1 & 0\\ 0 & 1 & -1 & 0\\ 0 & 0 & 0 & 3\end{pmatrix}$

$r(A)=2\quad r(\overline{A})=3$

$\because r(A)<r(\overline{A})\quad\therefore$ 无解

② $\lambda=1$

$\overline{A}=\begin{pmatrix}4 & 1 & 2 & 1\\ 1 & 0 & 1 & 1\\ 6 & 1 & 4 & 3\end{pmatrix}\to\begin{pmatrix}1 & 0 & 1 & 1\\ 0 & 1 & -2 & -3\\ 0 & 0 & 0 & 0\end{pmatrix}$

$\because r(A)=r(\overline{A})=2<3\quad\therefore$ 有无穷多解

③ $\lambda\neq0$ 或 1，即 $|A|\neq0\Rightarrow A$ 满秩，即 $r(A)=3$

$\because \overline{A}_{3\times4}\quad\therefore r(\overline{A})=3\quad\therefore r(A)=r(\overline{A})=3$，有唯一解

5. 已知 5 元线性方程组的系数矩阵的秩为 3, 且以下向量是它的解:

$$\boldsymbol{X}_1=(1,1,1,1,1),\quad \boldsymbol{X}_2=(1,2,3,4,5),\quad \boldsymbol{X}_3=(1,0,-3,-2,-3).$$

(1) 求方程组的通解.

(2) $\boldsymbol{X}_1+\boldsymbol{X}_2+\boldsymbol{X}_3$ 是否是方程组的解?

(3) $\frac{1}{3}(\boldsymbol{X}_1+\boldsymbol{X}_2+\boldsymbol{X}_3)$ 是否是方程组的解?

$A(X_1+X_2+X_3)=3b$

(1) 基础解系 $s=n-r(A)=2$

$[X_1\ X_2\ X_3]$

$$=\begin{pmatrix}1&1&1&1&1\\1&2&3&4&5\\1&0&-3&-2&-3\end{pmatrix}\to\begin{pmatrix}1&1&1&1&1\\0&1&2&3&4\\0&-1&-4&-3&-4\end{pmatrix}$$

$$\to\begin{pmatrix}1&1&1&1&1\\0&1&2&3&4\\0&0&-2&0&0\end{pmatrix}$$

$r=3$

$\therefore X_1、X_2、X_3$ 线性无关

若原方程为齐次方程, 则

$x=\eta_1 x_a+\eta_2 x_b$, $X_1、X_2、X_3$ 必线性相关

$\therefore$ 原方程非齐次, 记为 $Ax=b$

$\because X_1、X_2、X_3$ 为 $Ax=b$ 的特解

则 $X_2-X_1=(0,1,2,3,4)$ 为 $Ax=0$ 的解

$X_1-X_3=(0,1,4,3,4)$

$\because X_2-X_1,\ X_1-X_3$ 线性无关

$$\therefore x=\begin{pmatrix}0\\1\\2\\3\\4\end{pmatrix}t_1+\begin{pmatrix}0\\1\\4\\3\\4\end{pmatrix}t_2+\begin{pmatrix}1\\1\\1\\1\\1\end{pmatrix}$$

$$A=\begin{pmatrix} n & -1 & \cdots & -1 \\ -1 & n & \cdots & -1 \\ \vdots & \vdots & \ddots & \vdots \\ -1 & -1 & \cdots & n \end{pmatrix} \quad \text{求} A^{-1}$$

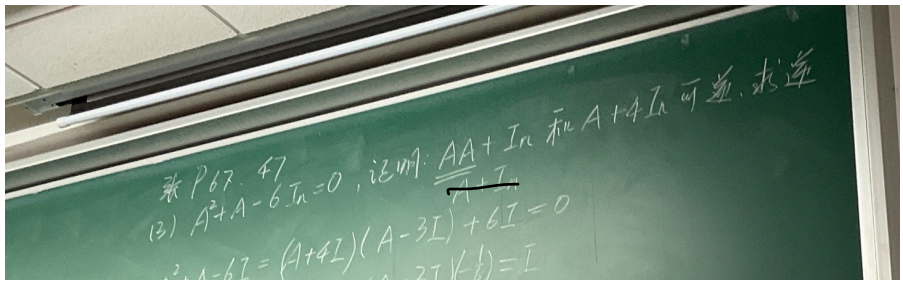
$$\left(\begin{array}{cccc|cccc} n & -1 & \cdots & -1 & 1 & & & \\ -1 & n & \cdots & -1 & & 1 & & \\ \vdots & \vdots & \ddots & \vdots & & & \ddots & \\ -1 & -1 & \cdots & n & & & & 1 \end{array}\right) \xrightarrow[i\geq 2]{R_i - R_1} \left(\begin{array}{cccc|cccc} n & -1 & \cdots & -1 & 1 & & & \\ -1-n & n+1 & \cdots & 0 & -1 & 1 & & \\ \vdots & \vdots & \ddots & \vdots & \vdots & & \ddots & \\ -1-n & 0 & \cdots & n+1 & -1 & & & 1 \end{array}\right)$$

$$\xrightarrow[i\geq 2]{R_1+\frac{1}{n+1}R_i} \left(\begin{array}{cccc|cccc} 1 & 0 & \cdots & 0 & \frac{2}{n+1} & \frac{1}{n+1} & \cdots & \frac{1}{n+1} \\ -(n+1) & n+1 & \cdots & 0 & -1 & 1 & & \\ \vdots & \vdots & \ddots & \vdots & \vdots & & \ddots & \\ -(n+1) & 0 & \cdots & n+1 & -1 & & & 1 \end{array}\right)$$

$$\xrightarrow[i\geq 2]{R_i+(n+1)R_1} \left(\begin{array}{cccc|cccc} 1 & & & & \frac{2}{n+1} & \frac{1}{n+1} & \cdots & \frac{1}{n+1} \\ & n+1 & & & 1 & 2 & \cdots & 1 \\ & & \ddots & & \vdots & \vdots & \ddots & \vdots \\ & & & n+1 & 1 & 1 & \cdots & 2 \end{array}\right)$$

$$\xrightarrow[i\geq 2]{R_i\times\frac{1}{n+1}} \left(\begin{array}{cccc|cccc} 1 & & & & \frac{2}{n+1} & \frac{1}{n+1} & \cdots & \frac{1}{n+1} \\ & 1 & & & \frac{1}{n+1} & \frac{2}{n+1} & \cdots & \frac{1}{n+1} \\ & & \ddots & & \vdots & \vdots & \ddots & \vdots \\ & & & 1 & \frac{1}{n+1} & \frac{1}{n+1} & \cdots & \frac{2}{n+1} \end{array}\right)$$

$$\therefore A^{-1}=\frac{1}{n+1}\begin{pmatrix} 2 & 1 & \cdots & 1 \\ 1 & 2 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 2 \end{pmatrix}$$



$A^2=-A+6I_n$

$A^4=A^2-12A+36I_n$

$(A^2+I)(A^2-14I)=-50I$

$(A^2+I)^{-1}=\frac{14I-A^2}{50}$

$A^2+A-6I_n=0$

$A^2+A=6I_n$

$$\begin{array}{r|l} & A-3 \\ \hline A+4 & A^2+A \\ & A^2+4A \\ \hline & -3A \\ & -3A-12 \\ \hline & 12 \end{array}$$

$(A+4I_n)(A-3I_n)+12I_n=6I_n$

$(A+4I_n)(A-3I_n)=-6I_n$

$A^2+I_n=-A+7I_n$

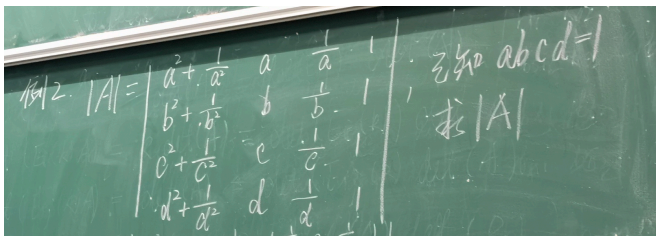
$A^2+A=6I_n$

$$\begin{array}{r|l} & A-8 \\ \hline -A+7 & A^2+A \\ & A^2-7A \\ \hline & 8A \\ & 8A-56 \\ \hline & 56 \end{array}$$

$(-A+7I_n)(-A-8I_n)+56I_n=6I_n$

$(-A+7I_n)(-A-8I_n)=-50I_n$

$(-A+7I_n)\frac{A+8I_n}{50}=I_n$



$$|A| = \begin{vmatrix} a^2 & a & \frac{1}{a} & 1 \\ b^2 & b & \frac{1}{b} & 1 \\ c^2 & c & \frac{1}{c} & 1 \\ d^2 & d & \frac{1}{d} & 1 \end{vmatrix} + \begin{vmatrix} \frac{1}{a^2} & a & \frac{1}{a} & 1 \\ \frac{1}{b^2} & b & \frac{1}{b} & 1 \\ \frac{1}{c^2} & c & \frac{1}{c} & 1 \\ \frac{1}{d^2} & d & \frac{1}{d} & 1 \end{vmatrix}$$

$$= -\begin{vmatrix} a^2 & a & 1 & \frac{1}{a} \\ b^2 & b & 1 & \frac{1}{b} \\ c^2 & c & 1 & \frac{1}{c} \\ d^2 & d & 1 & \frac{1}{d} \end{vmatrix} + \begin{vmatrix} a & 1 & \frac{1}{a} & \frac{1}{a^2} \\ b & 1 & \frac{1}{b} & \frac{1}{b^2} \\ c & 1 & \frac{1}{c} & \frac{1}{c^2} \\ d & 1 & \frac{1}{d} & \frac{1}{d^2} \end{vmatrix}$$

$$= -\begin{vmatrix} a^2 & a & 1 & \frac{1}{a} \\ b^2 & b & 1 & \frac{1}{b} \\ c^2 & c & 1 & \frac{1}{c} \\ d^2 & d & 1 & \frac{1}{d} \end{vmatrix} + \frac{1}{abcd}\begin{vmatrix} a^2 & a & 1 & \frac{1}{a} \\ b^2 & b & 1 & \frac{1}{b} \\ c^2 & c & 1 & \frac{1}{c} \\ d^2 & d & 1 & \frac{1}{d} \end{vmatrix}$$

$\because abcd = 1$

$\therefore$ 原式 $= 0$

已知n阶方阵为反对称阵，求证：n为奇时，$\det(A)=0$

证：$\det(A)=\det(A^T)$

$A=-A^T$

$\therefore \det(A)=\det(-A^T)=(-1)^n\det(A^T)=(-1)^n\det(A)$

若n奇，则$\det(A)=0$

A为代数余子式，M为余子式

$$\begin{vmatrix} 3 & -5 & 2 & 1 \\ 1 & 1 & 0 & -5 \\ -1 & 3 & 1 & 3 \\ 2 & -4 & -1 & -3 \end{vmatrix}$$

求 $A_{11}+A_{12}+A_{13}+A_{14}$

$M_{11}+M_{21}+\cdots+M_{41}$

半！

凑

$$\text{令：}|A_{11}|+|A_{12}|+|A_{13}|+|A_{14}|=\begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & -5 \\ -1 & 3 & 1 & 3 \\ 2 & -4 & -1 & -3 \end{vmatrix}$$

$$M_{11}+M_{21}+M_{31}+M_{41}=A_{11}-A_{21}+A_{31}-A_{41}=\begin{vmatrix} 1 & -5 & 2 & 1 \\ -1 & 1 & 0 & -5 \\ 1 & 3 & 1 & 3 \\ -1 & -4 & -1 & -3 \end{vmatrix}$$

$$D=\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ b+c & a+c & a+b \end{vmatrix}$$

$$=\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ a+b+c & a+b+c & a+b+c \end{vmatrix}$$

$$=(a+b+c)\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ 1 & 1 & 1 \end{vmatrix}$$

$$|A|=\begin{vmatrix} a & b & b & \cdots & b \\ b & a & b & \cdots & b \\ b & b & a & \cdots & b \\ \vdots & \vdots & \vdots & & \vdots \\ b & b & b & \cdots & a \end{vmatrix}_{n\times n}=\begin{vmatrix} a+(n-1)b & b & b & \cdots & b \\ a+(n-1)b & a & b & \cdots & b \\ a+(n-1)b & b & a & \cdots & b \\ \vdots & \vdots & \vdots & & \vdots \\ a+(n-1)b & b & b & \cdots & a \end{vmatrix}$$

$$=[a+(n-1)b]\begin{vmatrix} 1 & b & b & \cdots & b \\ 1 & a & b & \cdots & b \\ 1 & b & a & \cdots & b \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & b & b & \cdots & a \end{vmatrix}$$

(-1, -1, -1)

$$=[a+(n-1)b]\begin{vmatrix} 1 & b & b & \cdots & b \\ 0 & a-b & 0 & \cdots & 0 \\ 0 & 0 & a-b & \cdots & 0 \\ \vdots & 0 & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & a-b \end{vmatrix}$$

$$=[a+(n-1)b](a-b)^{n-1}$$

$$D=\begin{vmatrix} x_1+a & a & \cdots & a & a \\ a & x_2+a & \cdots & a & a \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ a & a & \cdots & x_{n-1}+a & a \\ a & a & \cdots & a & a \end{vmatrix}=\begin{vmatrix} x_1 & & & & a \\ & x_2 & \cdots & 0 & a \\ & \vdots & & \vdots & \vdots \\ & 0 & \cdots & x_{n-1} & a \\ & & & & a \end{vmatrix}=a x_1 x_2 \cdots x_{n-1}$$

$$D=\begin{vmatrix} x_1+a_1 & a_2 & a_3 & \cdots & a_n \\ a_1 & x_2+a_2 & a_3 & \cdots & a_n \\ \vdots & \vdots & x_3+a_3 & \cdots & a_n \\ a_1 & a_2 & a_3 & \cdots & x_n+a_n \end{vmatrix}$$ 加边法

$$=\begin{vmatrix} 1 & -a_1 & -a_2 & -a_3 & \cdots & -a_n \\ 0 & x_1+a_1 & a_2 & a_3 & \cdots & a_n \\ 0 & a_1 & x_2+a_2 & a_3 & \cdots & a_n \\ \vdots & \vdots & \vdots & x_3+a_3 & \cdots & a_n \\ 0 & a_1 & a_2 & a_3 & \cdots & x_n+a_n \end{vmatrix}$$

$$=\begin{vmatrix} 1 & a_1 & a_2 & a_3 & \cdots & a_n \\ 1 & x_1 & 0 & 0 & \cdots & 0 \\ 1 & 0 & x_2 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots \\ 1 & 0 & 0 & 0 & \cdots & x_n \end{vmatrix}$$

$$=x_1 x_2 \cdots x_n \begin{vmatrix} 1 & a_1 & a_2 & \cdots & a_n \\ \frac{1}{x_1} & 1 & 0 & \cdots & 0 \\ \frac{1}{x_2} & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & & \vdots \end{vmatrix}=\prod_{i=1}^{n} x_i \begin{vmatrix} 1-\sum_{i=1}^{n}\frac{a_i}{x_i} & 0 & 0 & \cdots & 0 \\ \frac{1}{x_1} & 1 & 0 & \cdots & 0 \\ \frac{1}{x_2} & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \end{vmatrix}$$

$$\begin{vmatrix} \vdots & \vdots & \vdots & & \vdots \\ \frac{1}{x_n} & 0 & 0 & \cdots & 1 \end{vmatrix} \qquad \begin{vmatrix} \vdots & \vdots & \vdots & & \vdots \\ \frac{1}{x_n} & 0 & 0 & \cdots & 1 \end{vmatrix}$$

$$= \prod_{i=1}^{n} x_i \left(1 - \sum_{i=1}^{n} \frac{a}{x_i}\right)$$

$$D_{2n} = \begin{vmatrix} x & 0 & \cdots & 0 & y \\ 0 & x & \cdots & y & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & y & \cdots & x & 0 \\ y & 0 & \cdots & 0 & x \end{vmatrix}$$

$$= \begin{vmatrix} x+y & 0 & \cdots & 0 & x+y \\ 0 & x+y & \cdots & x+y & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & y & \cdots & x & 0 \\ y & 0 & \cdots & 0 & x \end{vmatrix}$$

$$= (x+y)^n \begin{vmatrix} 1 & 0 & \cdots & 0 & 1 \\ 0 & 1 & \cdots & 1 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & y & \cdots & x & 0 \\ y & 0 & \cdots & 0 & x \end{vmatrix} \begin{matrix} \\ \\ \\ -y \\ -y \end{matrix}$$

$$= (x+y)^n \begin{vmatrix} 1 & 0 & \cdots & 0 & 1 \\ 0 & 1 & \cdots & 1 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & x-y & 0 \\ 0 & 0 & \cdots & 0 & x-y \end{vmatrix}$$

$$= (x+y)^n (x-y)^n$$

$$D_n=\begin{vmatrix} x & y & 0 & \cdots & 0 & 0 \\ 0 & x & y & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & x & y \\ y & 0 & 0 & \cdots & 0 & x \end{vmatrix}$$

$$D_n=\begin{vmatrix} x+y & xy & 0 & \cdots & 0 & 0 & 0 \\ 1 & x+y & xy & \cdots & 0 & 0 & 0 \\ 0 & 1 & x+y & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 & x+y \end{vmatrix}$$

$D_n = x^n + (-1)^{n+1} y^n$

$D_n = (x+y)D_{n-1} - xyD_{n-2}$

$D^2 = (x+y)D - xy$

$D^2 - (x+y)D + xy = 0$

$(D-x)(D-y) = 0$

$D = x$ 或 $D = y$

$D = \frac{x^{n-1} - y^{n-1}}{x-y}$

$$D_n=\begin{vmatrix} 2 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 1 & 2 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 2 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 1 & 2 \end{vmatrix}$$

$= 2D_{n-1} - D_{n-2}$

$x^2 = 2x - 1$

$x^2 - 2x + 1 = 0$

$(x-1)^2 = 0$

$x = 1$

$D_n = A + Bn$

$\begin{cases} D_1 = A + B = 2 \\ D_2 = A + 2B = 3 \end{cases}$

$\therefore A = 1$

$B = 1$

$\therefore D_n = n + 1$

$x^2 - 2\cos\alpha \cdot x + 1 = 0$

$\Delta = 4\cos^2\alpha - 4$

$= 4(\cos^2\alpha - 1)$

$\frac{2\cos\alpha \pm 2i\sin\alpha}{2} = \cos\alpha \pm i\sin\alpha = e^{\pm i\alpha}$

$$D_n = \begin{vmatrix} 2\cos\theta & 1 & 0 & \cdots & 0 & 0 & 0 \\ 1 & 2\cos\theta & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 2\cos\theta & 1 \\ 0 & 0 & 0 & \cdots & 0 & 1 & 2\cos\theta \end{vmatrix}$$

$D_n = 2\cos\theta D_{n-1} - D_{n-2}$

$x^2 = 2\cos\theta x - 1$

$x^2 - 2\cos\theta x + 1 = 0$

① $\cos\theta = 1$ $x_1 = x_2 = 1$

$D_n = A + Bn$

$D_1 = A + B = 2$ $D_2 = A + 2B = 3$ $\begin{cases} A = 1 \\ B = 1 \end{cases}$

$\therefore D = n + 1$

② $\cos\theta < 1$ $\Delta < 0$ 无解

$D_1 = 2\cos\theta$

$D_2 = 4\cos^2\theta - 1$

$D_3 = 4\cos\theta(2\cos\theta - 1)$

$\frac{\sin 2\theta}{\sin\theta} - 1 = \frac{\sin 2\theta - \sin\theta}{\sin\theta}$

$= \frac{(\sin 2\theta - \sin\theta)(\sin 2\theta + \sin\theta)}{\sin^2\theta}$

$= \frac{2\cos\frac{3\theta}{2}\sin\frac{\theta}{2} \cdot 2\sin\frac{3\theta}{2}\cos\frac{\theta}{2}}{\sin^2\theta}$

$= \frac{\sin 3\theta}{\sin\theta}$

② $2\cos\theta = \frac{\sin 2\theta}{\sin\theta}$

$D_1 = \frac{\sin 2\theta}{\sin\theta}$

$D_2 = 4\cos^2\theta - 1 = \frac{\sin 3\theta}{\sin\theta}$

猜想 $D_n = \frac{\sin(n+1)\theta}{\sin\theta}$

$n = 1, 2$ 时 ✓

假设 $n < k$ ✓

$n = k$ 时

$D_n = 2\cos\theta D_{n-1} - D_{n-2}$

$= 2\cos\theta \frac{\sin n\theta}{\sin\theta} - \frac{\sin(n-1)\theta}{\sin\theta}$

$= \frac{\sin(n+1)\theta + \sin(n-1)\theta - \sin(n-1)\theta}{\sin\theta}$

$= \frac{\sin(n+1)\theta}{\sin\theta}$

假设成立

$\frac{1}{x}$ $\frac{1}{x}$ $\frac{1}{x}$

$$\begin{vmatrix} x & -1 & 0 & \cdots & 0 & 0 \\ 0 & x & -1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & x & -1 \\ a_n & a_{n-1} & a_{n-2} & \cdots & a_2 & a_1+x \end{vmatrix}$$

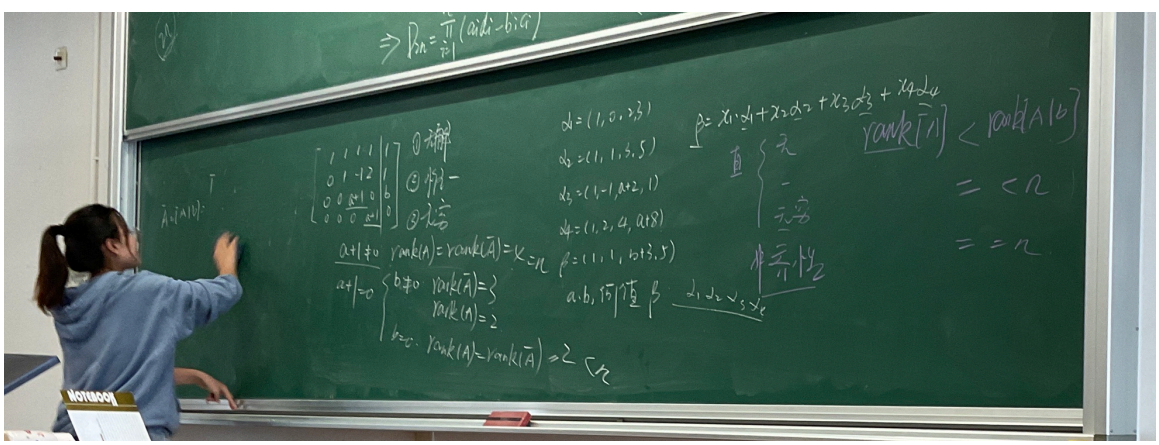
$$\left(\sum_{i=1}^{n} \frac{a_i}{x^{i-1}} + x\right) x^{n-1}$$

$$\begin{vmatrix} a_n & & & & b_n \\ & \ddots & & \iddots & \\ & & a_1\ b_1 & & \\ & & c_1\ d_1 & & \\ & \iddots & & \ddots & \\ c_n & & & & d_n \end{vmatrix}$$

① 行变换

$$\begin{vmatrix} a_n - c_n\cdot\frac{b_n}{d_n} & & & & \\ & \ddots & & & \\ & & c_1 - c_1\frac{d_1}{b_1} & & \\ & & & \ddots & \\ & & & & c_n \end{vmatrix} \begin{vmatrix} & & & & * \\ & & & \iddots & \\ & & c_1 & & \\ & \iddots & & \ddots & \\ 0 & & & & d_n \end{vmatrix} = \prod_{i}^{n} \left(a_i - c_i\frac{b_i}{d_i}\right) d_i$$

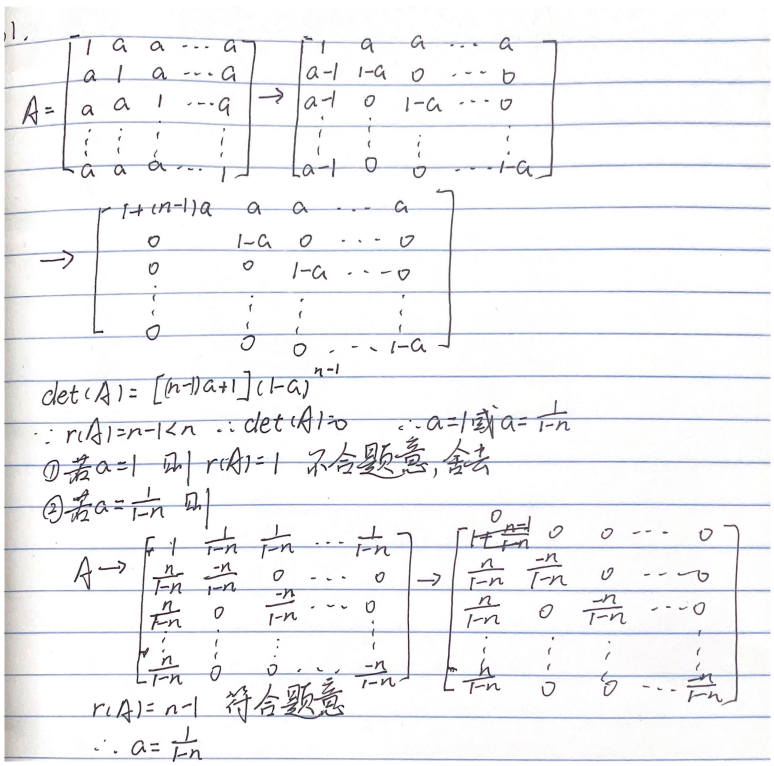
$$= \prod_{i}^{n} (a_i d_i - c_i b_i)$$

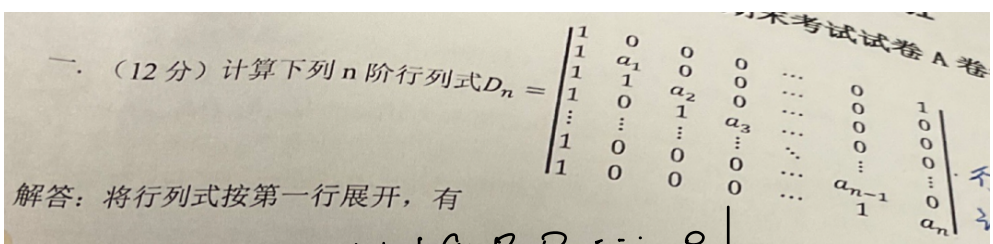


31. 设 $n(n \geqslant 3)$ 阶矩阵

$$\boldsymbol{A}=\begin{bmatrix} 1 & a & a & \cdots & a \\ a & 1 & a & \cdots & a \\ a & a & 1 & \cdots & a \\ \vdots & \vdots & \vdots & & \vdots \\ a & a & a & \cdots & 1 \end{bmatrix}$$

若矩阵 $\boldsymbol{A}$ 的秩为 $n-1$, 求 a.



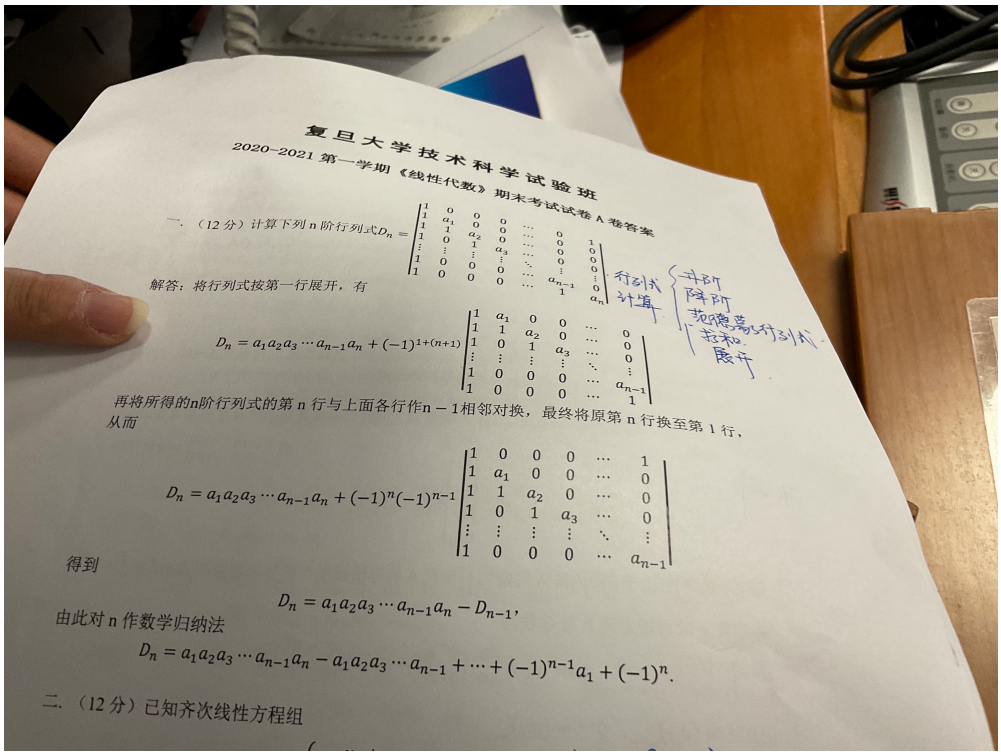


$$D_n=\prod_{i=1}^{n} a_i+(-1)^{n+1}\begin{vmatrix} 1 & a_1 & 0 & 0 & \cdots & 0 \\ 1 & 1 & a_2 & 0 & \cdots & 0 \\ 1 & 0 & 1 & a_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & 0 & \cdots & a_{n-1} \\ 1 & 0 & 0 & 0 & \cdots & 1 \end{vmatrix}$$

D_{n-1}

$$D_{n-1}=(-1)^{n+1}\prod_{i=1}^{n-1} a_i+D_{n-2}=(-1)^{n+1}\prod_{i=1}^{n-1} a_i+(-1)^{n}\prod_{i=1}^{n-2} a_i+\cdots+(-1)^{4}\prod_{i=1}^{2} a_i+(-1)^{3}a_1$$

$$\therefore D_n=\prod_{i=1}^{n} a_i+\prod_{i=1}^{n-1} a_i-\prod_{i=1}^{n-2} a_i+\prod_{i=1}^{n-3} a_i-\cdots+(-1)^{n+5}a_1a_2+(-1)^{n+4}a_1$$



复旦大学技术科学试验班

2020–2021 第一学期《线性代数》期末考试试卷 A 卷答案

一.（12 分）计算下列 n 阶行列式 $D_n=\begin{vmatrix} 1 & 0 & 0 & 0 & \cdots & 0 & 1 \\ 1 & a_1 & 0 & 0 & \cdots & 0 & 0 \\ 1 & 1 & a_2 & 0 & \cdots & 0 & 0 \\ 1 & 0 & 1 & a_3 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & 0 & 0 & 0 & \cdots & a_{n-1} & 0 \\ 1 & 0 & 0 & 0 & \cdots & 1 & a_n \end{vmatrix}$

解答：将行列式按第一行展开，有

$$D_n=a_1a_2a_3\cdots a_{n-1}a_n+(-1)^{1+(n+1)}\begin{vmatrix} 1 & a_1 & 0 & 0 & \cdots & 0 \\ 1 & 1 & a_2 & 0 & \cdots & 0 \\ 1 & 0 & 1 & a_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & 0 & \cdots & a_{n-1} \\ 1 & 0 & 0 & 0 & \cdots & 1 \end{vmatrix}$$

再将所得的n阶行列式的第 n 行与上面各行作 $n-1$ 相邻对换，最终将原第 n 行换至第 1 行，从而

$$D_n=a_1a_2a_3\cdots a_{n-1}a_n+(-1)^n(-1)^{n-1}\begin{vmatrix} 1 & 0 & 0 & 0 & \cdots & 1 \\ 1 & a_1 & 0 & 0 & \cdots & 0 \\ 1 & 1 & a_2 & 0 & \cdots & 0 \\ 1 & 0 & 1 & a_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & 0 & \cdots & a_{n-1} \end{vmatrix}$$

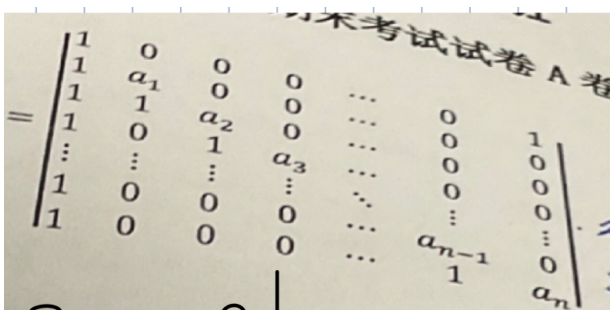
得到

$$D_n=a_1a_2a_3\cdots a_{n-1}a_n-D_{n-1},$$

由此对 n 作数学归纳法

$$D_n=a_1a_2a_3\cdots a_{n-1}a_n-a_1a_2a_3\cdots a_{n-1}+\cdots+(-1)^{n-1}a_1+(-1)^n.$$

二.（12 分）已知齐次线性方程组



$$D_{n+1}=\prod_{i=1}^{n}a_i+(-1)^{n+1}\begin{vmatrix} 1 & a_1 & 0 & 0 & \cdots & 0 \\ 1 & 1 & a_2 & 0 & \cdots & 0 \\ 1 & 0 & 1 & a_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & 0 & \cdots & a_{n-1} \\ 1 & 0 & 0 & 0 & \cdots & 1 \end{vmatrix}$$

$$=\prod_{i=1}^{n}a_i-D_{n-1}$$

$$=\prod_{i=1}^{n}a_i+\prod_{i=1}^{n-1}a_i+\prod_{i=1}^{n-2}a_i+\cdots+\prod_{i=1}^{1}a_i+D_0 \quad (D_0=a_1)$$

$$=\prod_{i=1}^{n}a_i-\prod_{i=1}^{n-1}a_i+\prod_{i=1}^{n-2}a_i+\cdots+(-1)^{n-1}a_1+(-1)^n$$

13. 计算 n 阶行列式：

$$\begin{vmatrix} 1 & 2 & 3 & \cdots & n \\ 2 & 3 & 4 & \cdots & 1 \\ 3 & 4 & 5 & \cdots & 2 \\ \vdots & \vdots & \vdots & & \vdots \\ n & 1 & 2 & \cdots & n-1 \end{vmatrix}$$

14. 计算 n 阶行列式：

13. 已知 4 阶矩阵 $\boldsymbol{A}=[\boldsymbol{\alpha}_1,\boldsymbol{\alpha}_2,\boldsymbol{\alpha}_3,\boldsymbol{\alpha}_4]$, $\boldsymbol{\alpha}_1,\boldsymbol{\alpha}_2,\boldsymbol{\alpha}_3,\boldsymbol{\alpha}_4$ 均为 4 元列向量, 其中 $\boldsymbol{\alpha}_2,\boldsymbol{\alpha}_3,\boldsymbol{\alpha}_4$ 线性无关, $\boldsymbol{\alpha}_1=2\boldsymbol{\alpha}_2-\boldsymbol{\alpha}_3$, 若 $\boldsymbol{\beta}=\boldsymbol{\alpha}_1+\boldsymbol{\alpha}_2+\boldsymbol{\alpha}_3+\boldsymbol{\alpha}_4$, 求线性方程组 $\boldsymbol{Ax}=\boldsymbol{\beta}$ 的通解.

$\because \beta=\alpha_1+\alpha_2+\alpha_3+\alpha_4$

$\therefore Ax=\beta$ 有解

$\because \alpha_2、\alpha_3、\alpha_4$ 线无，$\alpha_1=2\alpha_2-\alpha_3$

$\therefore rank(A)=3$

$\therefore$ 有无穷多解　基础解系向量数 $s=n-rank(A)=1$

$\because (\alpha_1\ \alpha_2\ \alpha_3\ \alpha_4)\begin{pmatrix}1\\1\\1\\1\end{pmatrix}=\alpha_1+\alpha_2+\alpha_3+\alpha_4=\beta$

$\therefore \begin{pmatrix}1\\1\\1\\1\end{pmatrix}$ 为一 $Ax=\beta$ 特解

$\because (\alpha_1\ \alpha_2\ \alpha_3\ \alpha_4)\begin{pmatrix}1\\-2\\1\\0\end{pmatrix}=\alpha_1-2\alpha_2+\alpha_3=0$

$\therefore \begin{pmatrix}1\\-2\\1\\0\end{pmatrix}$ 为 $Ax=0$ 一非零解

$\therefore x=\begin{pmatrix}1\\1\\1\\1\end{pmatrix}+\begin{pmatrix}1\\-2\\1\\0\end{pmatrix}t\quad(t\in R)$

线空 R^2, $\alpha=\begin{bmatrix}3\\4\end{bmatrix}$ $\beta=\begin{bmatrix}-1\\7\end{bmatrix}$

(1) 求 $\alpha\beta$ 距离 (2) 将 $\alpha\beta$ 单位化 (3) 夹角

解: (1) $d=5$

(2) $\|\alpha\|=5$ $\alpha_E=\begin{bmatrix}\frac{3}{5}\\\frac{4}{5}\end{bmatrix}$

$\|\beta\|=5\sqrt{2}$ $\beta_E=\begin{bmatrix}-\frac{1}{5\sqrt{2}}\\\frac{7}{5\sqrt{2}}\end{bmatrix}$

(3) $\cos\theta=\frac{\langle\alpha,\beta\rangle}{\|\alpha\|\|\beta\|}=\frac{-3+28}{5\times5\sqrt{2}}=\frac{\sqrt{2}}{2}$

$\therefore\theta=45^\circ$

求 $\alpha=[1,2,2,3]$ $\beta=[3\ 1\ 5\ 1]$ 夹角

$\cos\theta=\frac{\langle\alpha,\beta\rangle}{\|\alpha\|\|\beta\|}=\frac{3+2+10+3}{3\sqrt{2}\times6}=\frac{\sqrt{2}}{2}$

$\therefore\theta=\frac{\pi}{4}$

$f(x)=\sin x$ $g(x)=\cos x$ $[-\pi,\pi]$ 是否正交、长度?

$\langle f\ g\rangle=\int_{-\pi}^{\pi}\sin x\cos x\,dx$

$=\frac{1}{2}\int_{-\pi}^{\pi}\sin 2x\,dx$

$=\frac{1}{2}\left[-\frac{1}{2}\cos 2x\right]_{-\pi}^{\pi}$

$=-\frac{1}{4}[1-1]$

$=0$ ✓

$\|f\|^2=\langle f,f\rangle$

$=\int_{-\pi}^{\pi}\sin^2 x\,dx$

$=\int_{-\pi}^{\pi}\frac{1-\cos 2x}{2}dx$

$=\int_{-\pi}^{\pi}\left(\frac{1}{2}-\frac{1}{2}\cos 2x\right)dx$

$=\left[\frac{1}{2}x-\frac{1}{4}\sin 2x\right]_{-\pi}^{\pi}$

$=\frac{1}{2}\pi+\frac{1}{2}\pi$

$=\pi$

$\therefore\|f\|=\sqrt{\pi}$

$g(x) = \cos 3x$

$\|g\|^2 = \langle g, g \rangle$

$= \int_{-\pi}^{\pi} \cos^2 x \, dx$

$= \int_{-\pi}^{\pi} \frac{1+\cos 2x}{2} \, dx$

$= \left[\frac{1}{2}x + \frac{1}{4}\sin 2x \right]_{-\pi}^{\pi}$

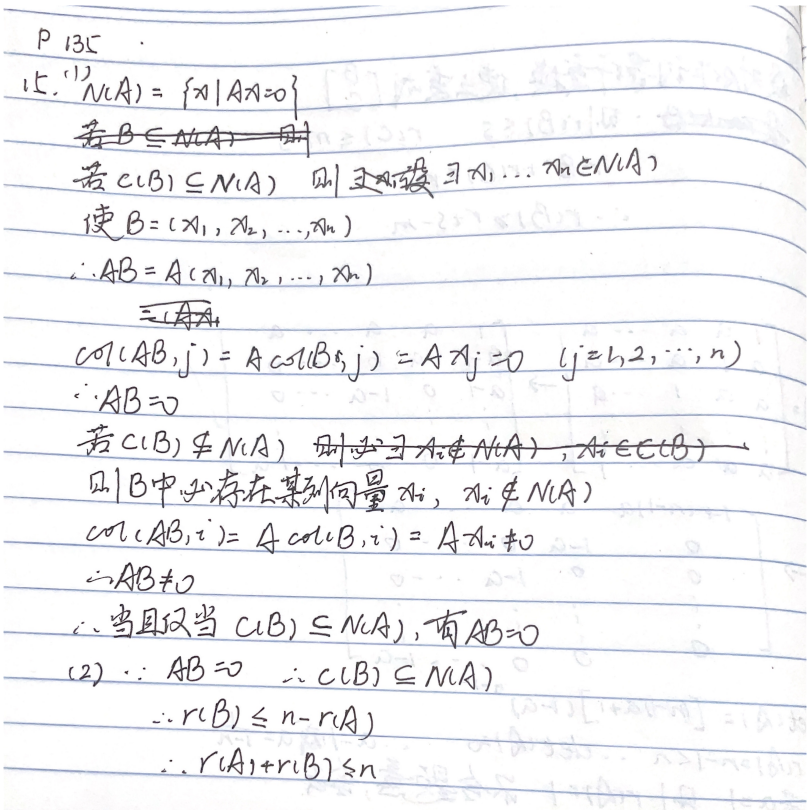
$= \pi$

$\therefore \|g\| = \sqrt{\pi}$

15. 令 $\boldsymbol{A}$ 和 $\boldsymbol{B}$ 为 $n \times n$ 矩阵.

(1) 证明: 当且仅当 $\boldsymbol{B}$ 的列空间是 $\boldsymbol{A}$ 的零空间的子空间时, 有 $\boldsymbol{AB}=\boldsymbol{O}$;

(2) 证明: 若 $\boldsymbol{AB}=\boldsymbol{O}$, 则 $\boldsymbol{A}$ 和 $\boldsymbol{B}$ 的秩的和不会超过 n.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证. (1) $N(A)=\{x \mid Ax=0\}$

~~若 $B \subseteq N(A)$ 则~~

若 $C(B) \subseteq N(A)$ 则 设 $x_1, \dots x_n \in N(A)$

使 $B=(x_1, x_2, \dots, x_n)$

$\therefore AB=A(x_1, x_2, \dots, x_n)$

~~$=(Ax_1)$~~

$col(AB, j)=A\,col(B, j)=Ax_j=0 \quad (j=1,2,\cdots,n)$

$\therefore AB=0$

若 $C(B) \not\subseteq N(A)$ ~~则 至少有 $x_i \notin N(A)$, $x_i \in C(B)$~~

则 B 中 至少存在某列向量 x_i, $x_i \notin N(A)$

$col(AB, i)=A\,col(B, i)=Ax_i \neq 0$

$\therefore AB \neq 0$

$\therefore$ 当且仅当 $C(B) \subseteq N(A)$, 有 $AB=0$

(2) $\because AB=0 \quad \therefore C(B) \subseteq N(A)$

$\therefore r(B) \leq n-r(A)$

$\therefore r(A)+r(B) \leq n$

特征值

12. 设非零向量 $\boldsymbol{\alpha}=[a_1,a_2,\cdots,a_n]^{\mathrm{T}}$, $\boldsymbol{\beta}=[b_1,b_2,\cdots,b_n]^{\mathrm{T}}$, 且 $\boldsymbol{\alpha}^{\mathrm{T}}\boldsymbol{\beta}=0$, $\boldsymbol{A}=\boldsymbol{\alpha}\boldsymbol{\beta}^{\mathrm{T}}$, 求 $\boldsymbol{A}$ 的特征值和特征向量.

$\because A=\alpha\beta^T$

$\therefore A^k=\alpha\beta^T\alpha\beta^T\cdots\alpha\beta^T=\alpha(\alpha^T\beta)^T(\alpha^T\beta)^T\cdots(\alpha^T\beta)^T\beta^T=\vec{0}$

$|\lambda I-A^k|=|\lambda I|=\lambda^n|I|=0$ 得 $\lambda_{k1}=\lambda_{k2}=\cdots=\lambda_{kn}=0$

$\because \lambda(A^k)=\lambda^k(A)$ $\therefore \lambda_1=\lambda_2=\cdots=\lambda_n=0$

$$A=\alpha\beta^T=\begin{bmatrix}a_1b_1 & a_1b_2 & \cdots & a_1b_n\\ a_2b_1 & a_2b_2 & \cdots & a_2b_n\\ \vdots & \vdots & & \vdots\\ a_nb_1 & a_nb_2 & \cdots & a_nb_n\end{bmatrix}$$

$$(\lambda I-A)=\begin{bmatrix}-a_1b_1 & -a_1b_2 & \cdots & -a_1b_n\\ -a_2b_1 & -a_2b_2 & \cdots & -a_2b_n\\ \vdots & \vdots & & \vdots\\ -a_nb_1 & -a_nb_2 & \cdots & -a_nb_n\end{bmatrix}\to\begin{bmatrix}-a_1b_1 & -a_1b_2 & \cdots & -a_1b_n\\ 0 & 0 & \cdots & 0\\ \vdots & \vdots & & \vdots\\ 0 & 0 & \cdots & 0\end{bmatrix}\to\begin{bmatrix}b_1 & b_2 & \cdots & b_n\\ 0 & 0 & \cdots & 0\\ \vdots & \vdots & & \vdots\\ 0 & 0 & \cdots & 0\end{bmatrix}$$

基解 $\begin{bmatrix}-\frac{b_2}{b_1}\\1\\0\\\vdots\\0\end{bmatrix}\begin{bmatrix}-\frac{b_3}{b_1}\\0\\1\\0\\\vdots\\0\end{bmatrix}\cdots\begin{bmatrix}-\frac{b_n}{b_1}\\0\\\vdots\\0\\1\end{bmatrix}$

$\therefore$ 特征向量 $x=k_1\begin{bmatrix}-\frac{b_2}{b_1}\\1\\0\\\vdots\\0\end{bmatrix}+k_2\begin{bmatrix}-\frac{b_3}{b_1}\\0\\1\\0\\\vdots\\0\end{bmatrix}+\cdots+k_{n-1}\begin{bmatrix}-\frac{b_n}{b_1}\\0\\\vdots\\0\\1\end{bmatrix}$

$(k_1,\cdots,k_{n-1}\in R$ 且 $k_1\cdots k_{n-1}\neq 0)$